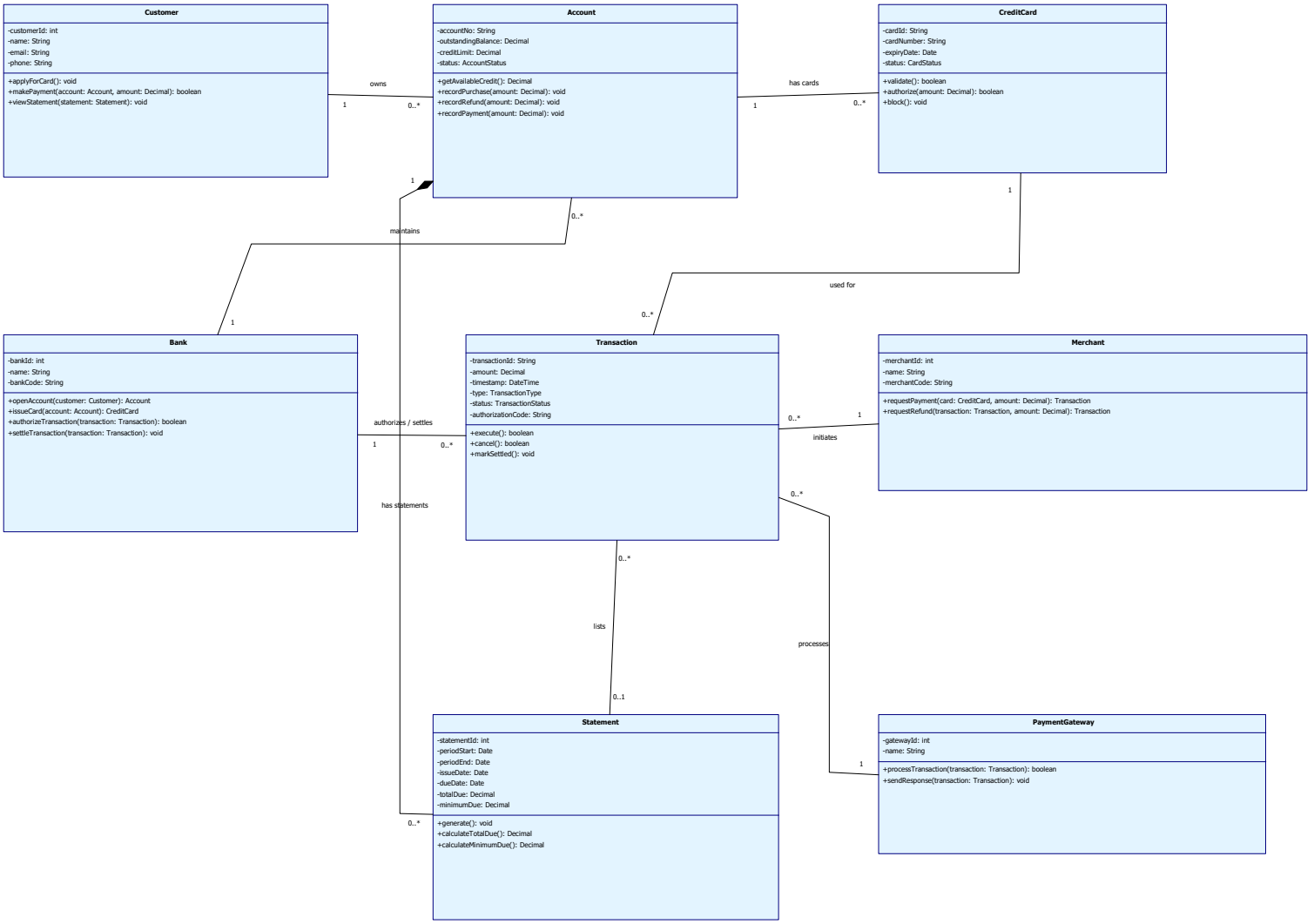

Customer
-customerId: int
-name: String
-email: String
-phone: String
+applyForCard(): void
+makePayment(account: Account, amount: Decimal): boolean
+viewStatement(statement: Statement): void
owns
1
0..*
Account
-accountNo: String
-outstandingBalance: Decimal
-creditLimit: Decimal
-status: AccountStatus
+getAvailableCredit(): Decimal
+recordPurchase(amount: Decimal): void
+recordRefund(amount: Decimal): void
+recordPayment(amount: Decimal): void
has cards
1
0..*
CreditCard
-cardId: String
-cardNumber: String
-expiryDate: Date
-status: CardStatus
+validate(): boolean
+authorize(amount: Decimal): boolean
+block(): void
1
maintains
0..*
1
used for
1
0..*
Bank
-bankId: int
-name: String
-bankCode: String
+openAccount(customer: Customer): Account
+issueCard(account: Account): CreditCard
+authorizeTransaction(transaction: Transaction): boolean
+settleTransaction(transaction: Transaction): void
authorizes / settles
1
0..*
Transaction
-transactionId: String
-amount: Decimal
-timestamp: DateTime
-type: TransactionType
-status: TransactionStatus
-authorizationCode: String
+execute(): boolean
+cancel(): boolean
+markSettled(): void
0..*
1
initiates
Merchant
-merchantId: int
-name: String
-merchantCode: String
+requestPayment(card: CreditCard, amount: Decimal): Transaction
+requestRefund(transaction: Transaction, amount: Decimal): Transaction
has statements
0..*
0..*
lists
0..1
processes
1
Statement
-statementId: int
-periodStart: Date
-periodEnd: Date
-issueDate: Date
-dueDate: Date
-totalDue: Decimal
-minimumDue: Decimal
+generate(): void
+calculateTotalDue(): Decimal
+calculateMinimumDue(): Decimal
0..*
PaymentGateway
-gatewayId: int
-name: String
+processTransaction(transaction: Transaction): boolean
+sendResponse(transaction: Transaction): void
<<enumeration>>
AccountStatus
+ACTIVE
+SUSPENDED
+CLOSED
<<enumeration>>
CardStatus
+ACTIVE
+BLOCKED
+EXPIRED
<<enumeration>>
TransactionType
+PURCHASE
+REFUND
<<enumeration>>
TransactionStatus
+PENDING
+AUTHORIZED
+DECLINED
+SETTLED
+CANCELLED